$$n = \frac{c}{v} \qquad (6.16)$$

Refractive index of a transparent medium gives an idea about the speed of light in that medium.

EXAMPLE 6.5

Pure water has refractive index 1.33. What is the speed of light through it?

Solution

$n = \frac{c}{v}$; $\quad v = \frac{c}{n}$

$$v = \frac{3\times10^8}{1.33} = 2.26\times10^8 \text{ m s}^{-1}$$

Light travels with a speed of $2.26\times10^8 \text{ m s}^{-1}$ through pure water.

Refractive index does not have a unit. The smallest value of refractive index is for vacuum, which is 1. For any other medium refractive index is greater than 1. Refractive index is also called as optical density of the medium. Higher the refractive index of a medium, lesser is the speed of light through it and vice-versa. [Note: optical density should not be confused with mass density of the material of the medium. They two are different entities]. The Table 6.2 shows the refractive indices of different transparent media.

Table 6.2 Refractive index of media

Media	Refractive index
Vacuum	1.00
Air	1.0003
Carbon dioxide gas	1.0005
Ice	1.31
Pure water	1.33
Ethyl alcohol	1.36
Quartz	1.46
Vegetable oil	1.47
Olive oil	1.48
Acrylic	1.49
Table salt	1.51
Glass	1.52
Sapphire	1.77
Zircon	1.92
Qubic zirconia	2.16
Diamond	2.42
Gallium phosphide	3.50

6.3.4 Optical path

***Optical path* of a medium is defined as the distance *d′* light travels in vacuum in the same time it travels a distance *d* in the medium.**

Let us consider a medium of refractive index n and thickness d. Light travels with a speed v through the medium in a time t. The speed of light through the medium is written as,

$$v = \frac{d}{t}; \text{ rewritten for } t \text{ as, } t = \frac{d}{v}$$

In the same time t, light can cover a longer distance d' in vacuum as it travels with greater speed c in vacuum. It is shown in Figure 6.14. Now, we can wrtite,

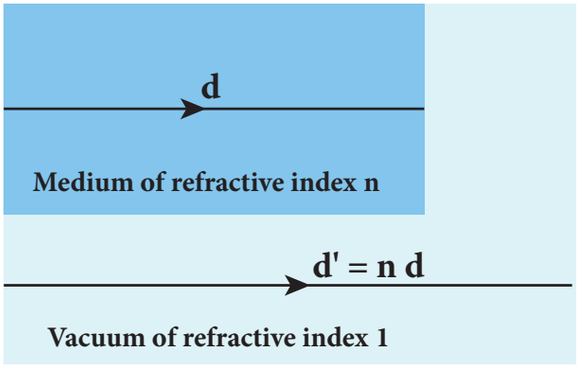


Figure 6.14 Optical path

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